

## Proportions of checks number

```
In [1]: import pandas as pd

def proportion_and_count_of_transactions_with_upsell(fileyear):
    # Construct the file path
    filepath = f"/Users/jingwenshi/VK-Lab Dropbox/Maria Shi/Data/VK_query_results/{fileyear}/clean.sales.detail.csv"
    # Read the CSV file into a DataFrame
    file2 = pd.read_csv(filepath)

    # Exclude "TakeAway Charge" from the item names if needed
    file2 = file2[file2['alohaitemname'] != "TakeAway Charge"]

    # Identify upsell transactions
    file2['is_upsell'] = file2['quickcomboupsell'] > 0

    # Group by 'businessdate', 'fkstoreid', and count unique transactions
    transaction_counts = file2.groupby(['businessdate', 'fkstoreid']).agg(total_transactions=('uid', 'nunique')).reset_index()

    # Count unique upsell transactions
    upsell_transactions = file2[file2['is_upsell']].groupby(['businessdate', 'fkstoreid']).agg(upsell_transactions=('uid', 'nunique')).reset_index()

    # Merge the counts to calculate proportions and include total upsell transactions
    merged_df = pd.merge(transaction_counts, upsell_transactions, on=['businessdate', 'fkstoreid'], how='left').fillna(0)
    merged_df['proportionUpsellTransactions'] = merged_df['upsell_transactions'] / merged_df['total_transactions']

    # The 'upsell_transactions' column already summarizes the total number of transactions with any upsell item
    # for each 'businessdate' and 'fkstoreid', so no further changes are needed for that part.

    # Select relevant columns for the final DataFrame
    result_df = merged_df[['businessdate', 'fkstoreid', 'total_transactions', 'upsell_transactions', 'proportionUpsellTransactions']]

    # Statistics for Proportion of Upsell Transactions
    print(fileyear)
    print(f"Mean Proportion: {result_df['proportionUpsellTransactions'].mean()}")
    print(f"Median Proportion: {result_df['proportionUpsellTransactions'].median()}")
    print(f"IQR Proportion: {result_df['proportionUpsellTransactions'].quantile(0.75) - result_df['proportionUpsellTransactions'].quantile(0.25)}")

    return result_df

fileyear = "Jan_2018"
pUpsellorder = proportion_and_count_of_transactions_with_upsell(fileyear)

lst = ["Feb_2018", "Mar_2018", "Apr_2018", "May_2018", "Jun_2018", "Jul_2018", "Aug_2018", "Sep_2018", "Oct_2018", "Nov_2018", "Dec_2018"]
for i in lst:
    dfToMerge = proportion_and_count_of_transactions_with_upsell(i)
    pUpsellorder = pd.concat([pUpsellorder, dfToMerge])
```

Jan\_2018  
 Mean Proportion: 0.06016423739188618  
 Median Proportion: 0.04885156125704429  
 IQR Proportion: 0.04074293166356776  
 Feb\_2018  
 Mean Proportion: 0.054661438398992256  
 Median Proportion: 0.04391891891891892  
 IQR Proportion: 0.03676528287170526  
 Mar\_2018  
 Mean Proportion: 0.0474357395949415  
 Median Proportion: 0.03833599574014909  
 IQR Proportion: 0.031493310090244774  
 Apr\_2018  
 Mean Proportion: 0.04965505049448621  
 Median Proportion: 0.04220779220779221  
 IQR Proportion: 0.03515595966603077  
 May\_2018  
 Mean Proportion: 0.04955648561769809  
 Median Proportion: 0.03834184525798291  
 IQR Proportion: 0.03516926326459899  
 Jun\_2018  
 Mean Proportion: 0.046958359799176966  
 Median Proportion: 0.0364741641337386  
 IQR Proportion: 0.03376466473786284  
 Jul\_2018  
 Mean Proportion: 0.06373427549725494  
 Median Proportion: 0.04534313725490196  
 IQR Proportion: 0.04271662085925068  
 Aug\_2018  
 Mean Proportion: 0.07528807122284324  
 Median Proportion: 0.05258549175039174  
 IQR Proportion: 0.0475134214664265  
 Sep\_2018  
 Mean Proportion: 0.09243386449907781  
 Median Proportion: 0.07191396028936432  
 IQR Proportion: 0.05652291989420941  
 Oct\_2018  
 Mean Proportion: 0.09285072056727375  
 Median Proportion: 0.07486275369394323  
 IQR Proportion: 0.04512761605972572  
 Nov\_2018  
 Mean Proportion: 0.12231173446574355  
 Median Proportion: 0.1  
 IQR Proportion: 0.07222084395227678  
 Dec\_2018  
 Mean Proportion: 0.1400083726359771  
 Median Proportion: 0.11567436892112218  
 IQR Proportion: 0.09831659634117837

```

In [2]: print(f"Mean: {pUpsellorder['proportionUpsellTransactions'].mean()}")
print(f"Median: {pUpsellorder['proportionUpsellTransactions'].median()}")
print(f"IQR: {pUpsellorder['proportionUpsellTransactions'].quantile(0.75) - pUpsellorder['proportionUpsellTransactions'].quantile(0.25)}")
print(f"SD: {pUpsellorder['proportionUpsellTransactions'].std()}")
print(pUpsellorder['proportionUpsellTransactions'].quantile([0.0, 0.25, 0.5, 0.75, 1.0]))
  
```

Mean: 0.07735802732570561  
Median: 0.05714285714285714  
IQR: 0.058186523262952566  
SD: 0.07054818599146204  
0.00 0.000000  
0.25 0.034562  
0.50 0.057143  
0.75 0.092749  
1.00 0.661721  
Name: proportionUpsellTransactions, dtype: float64

```
In [3]: # Number of Rows in the DataFrame
num_rows = len(pUpsellorder)

# Print the number of rows
print(f"Number of Rows: {num_rows}")
```

Number of Rows: 35581

## add the proportion of revenue

```
In [4]: def revenueProp(fileyear):
    filepath = f"/Users/jingwenshi/VK-Lab Dropbox/Maria Shi/Data/VK_query_results/{fileyear}/clean.sales.detail.csv"
    file2 = pd.read_csv(filepath)

    # Filtering out the TakeAway Charge
    file2 = file2[file2['alohaitemname'] != "TakeAway Charge"]

    # Calculating Total Price for upsell
    summedPrices = file2.groupby(['businessdate', 'fkstoreid', 'quickcomboupsell']).agg({'totalprice': 'sum'}).reset_index()
    summedPrices.rename(columns={'totalprice': 'upsell'}, inplace=True)

    # Summing upselling item revenue
    totalPrices = summedPrices.groupby(['businessdate', 'fkstoreid']).agg({'upsell': 'sum'}).reset_index()
    totalPrices.rename(columns={'upsell': 'totalprices'}, inplace=True)

    # Calculating Proportion
    newdf = pd.merge(totalPrices, summedPrices, on=['businessdate', 'fkstoreid'])
    newdf['proportionUpsellrevenue'] = newdf['upsell'] / newdf['totalprices']

    # Housekeeping the DataFrame
    newdf = newdf[newdf['quickcomboupsell'] == 1]
    newdf = newdf[['businessdate', 'fkstoreid', 'proportionUpsellrevenue']]

    # Tabulating statistics for Proportion of upsell Revenue
    print(fileyear)
    print(f"Mean: {newdf['proportionUpsellrevenue'].mean()}")
    print(f"Median: {newdf['proportionUpsellrevenue'].median()}")
    print(f"IQR: {newdf['proportionUpsellrevenue'].quantile(0.75) - newdf['proportionUpsellrevenue'].quantile(0.25)}")

    return newdf

fileyear = "Jan_2018"
pUpsellrevenue = revenueProp(fileyear)
```

```
lst = ["Feb_2018", "Mar_2018", "Apr_2018", "May_2018", "Jun_2018", "Jul_2018", "Aug_2018", "Sep_2018", "Oct_2018", "Nov_2018", "Dec_2018"]
for i in lst:
    dfToMerge = revenueProp(i)
    pUpsellrevenue = pd.concat([pUpsellrevenue, dfToMerge])
```

```
Jan_2018
Mean: 0.02751106499127068
Median: 0.02263571234723508
IQR: 0.01978518212283739
Feb_2018
Mean: 0.024780618381374048
Median: 0.019939597939507194
IQR: 0.01744382407476196
Mar_2018
Mean: 0.021784147943032815
Median: 0.018123729809238744
IQR: 0.014891244230988485
Apr_2018
Mean: 0.02317329483720363
Median: 0.019998086932075443
IQR: 0.016131616543326333
May_2018
Mean: 0.02246943783192419
Median: 0.0178634250681095
IQR: 0.01628546289666883
Jun_2018
Mean: 0.02040008447522971
Median: 0.015862549622882333
IQR: 0.014618280670289675
Jul_2018
Mean: 0.028917012063778644
Median: 0.0205288308935813
IQR: 0.01916304872067711
Aug_2018
Mean: 0.033711497012135036
Median: 0.02334884427404335
IQR: 0.02225464113175157
Sep_2018
Mean: 0.04162891563793094
Median: 0.03268437700266742
IQR: 0.02638316513196629
Oct_2018
Mean: 0.04380952021213265
Median: 0.03521434745613038
IQR: 0.022138774691646552
Nov_2018
Mean: 0.05603865621976071
Median: 0.04619697990255374
IQR: 0.03291705326048614
Dec_2018
Mean: 0.062024465168110596
Median: 0.05079454151265216
IQR: 0.045742759415441994
```

```
In [5]: print(f"Mean: {pUpsellrevenue['proportionUpsellrevenue'].mean()}")
print(f"Median: {pUpsellrevenue['proportionUpsellrevenue'].median()}")
print(f"IQR: {pUpsellrevenue['proportionUpsellrevenue'].quantile(0.75) - pUpsellrevenue['proportionUpsellrevenue'].quantile(0.25)}")
print(f"SD: {pUpsellrevenue['proportionUpsellrevenue'].std()}")
print(pUpsellrevenue['proportionUpsellrevenue'].quantile([0.0, 0.25, 0.5, 0.75, 1.0]))
```

```
Mean: 0.03510522602768723
Median: 0.026206130255482837
IQR: 0.02681869050200134
SD: 0.030859388504289376
0.00    0.000147
0.25    0.015754
0.50    0.026206
0.75    0.042572
1.00    0.254986
Name: proportionUpsellrevenue, dtype: float64
```

```
In [6]: # merge the proportion of quantity and revenue together
file = pd.merge(pUpsellorder, pUpsellrevenue)

# Displaying the first few rows of the merged DataFrame
print(file.head())

# Number of Rows in the DataFrame
num_rows = len(file)

# Print the number of rows
print(f"Number of Rows: {num_rows}")
```

|   | businessdate | fkstoreid | total_transactions | upsell_transactions \ |
|---|--------------|-----------|--------------------|-----------------------|
| 0 | 2018-01-01   | 16054.0   | 445                | 39.0                  |
| 1 | 2018-01-01   | 16227.0   | 600                | 27.0                  |
| 2 | 2018-01-01   | 16309.0   | 533                | 22.0                  |
| 3 | 2018-01-01   | 16702.0   | 524                | 16.0                  |
| 4 | 2018-01-01   | 17523.0   | 492                | 24.0                  |

|   | proportionUpsellTransactions | proportionUpsellrevenue |
|---|------------------------------|-------------------------|
| 0 | 0.087640                     | 0.038784                |
| 1 | 0.045000                     | 0.019313                |
| 2 | 0.041276                     | 0.018664                |
| 3 | 0.030534                     | 0.009755                |
| 4 | 0.048780                     | 0.023666                |

Number of Rows: 35538

```
In [7]: # Data Cleaning
file['businessdate'] = pd.to_datetime(file['businessdate'], format='%Y-%m-%d')

# Adding businessmonth, w.day, weeknum as factors
file['businessmonth'] = file['businessdate'].dt.month
file['w.day'] = file['businessdate'].dt.dayofweek
file['weeknum'] = ((file['businessdate'].dt.day - 1) // 7 + 1)

# Convert to categorical
file['businessmonth'] = file['businessmonth'].astype('category')
file['w.day'] = file['w.day'].astype('category')
```

```
file['weeknum'] = file['weeknum'].astype('category')

# Adding trend.var
file['trend.var'] = (file['businessdate'] - file['businessdate'].min()).dt.days + 1
```

```
In [8]: from datetime import datetime
import os

# Getting the public holidays data

path1 = '/Users/jingwenshi/VK-Lab Dropbox/Maria Shi/Data/rstudio-export-20230725/2018.public.holidays.indonesia.csv'
ph_list = pd.read_csv(path1)

# Housekeeping
ph_list['public_holiday'] = 1
ph_list['businessdate'] = pd.to_datetime(ph_list['date'], format='%m/%d/%y')
ph_list['ph_wday'] = ph_list['businessdate'].dt.day_name()
ph_list['ph_weekend'] = ph_list['public_holiday'] * ph_list['ph_wday'].isin(['Saturday', 'Sunday'])

# Selecting columns
tableToMerge = ph_list[['businessdate', 'public_holiday', 'ph_weekend']]

# Merging in PH Details
file_merged = pd.merge(file, tableToMerge, on='businessdate', how='left')

# Replacing NaN values with 0 in 'public_holiday' and 'ph_weekend'
file_merged['public_holiday'].fillna(0, inplace=True)
file_merged['ph_weekend'].fillna(0, inplace=True)

# Displaying the first few rows of the merged DataFrame
print(file_merged.head())
```

```
businessdate  fkstoreid  total_transactions  upsell_transactions  \
0  2018-01-01    16054.0             445             39.0
1  2018-01-01    16227.0             600             27.0
2  2018-01-01    16309.0             533             22.0
3  2018-01-01    16702.0             524             16.0
4  2018-01-01    17523.0             492             24.0

proportionUpsellTransactions  proportionUpsellrevenue  businessmonth  w.day  \
0             0.087640             0.038784             1             0
1             0.045000             0.019313             1             0
2             0.041276             0.018664             1             0
3             0.030534             0.009755             1             0
4             0.048780             0.023666             1             0

weeknum  trend.var  public_holiday  ph_weekend
0         1         1             1.0         0.0
1         1         1             1.0         0.0
2         1         1             1.0         0.0
3         1         1             1.0         0.0
4         1         1             1.0         0.0
```

```
In [9]: # Adding a dummy for the Ramadan period in 2018
file_merged['ramadan'] = 0
```

```

# Creating a mask for the Ramadan period
ramadan_start = '2018-05-16'
ramadan_end = '2018-06-14'
ramadan_mask = (file_merged['businessdate'] >= ramadan_start) & (file_merged['businessdate'] <= ramadan_end)

# Applying the mask to the DataFrame
file_merged.loc[ramadan_mask, 'ramadan'] = 1

# Optional: Creating a test DataFrame for Ramadan period
ramadan_test = file_merged[['businessdate', 'ramadan']].drop_duplicates()

# Displaying the first few rows of the merged DataFrame
print(file_merged.head())

```

```

businessdate  fkstoreid  total_transactions  upsell_transactions  \
0  2018-01-01    16054.0                445                39.0
1  2018-01-01    16227.0                600                27.0
2  2018-01-01    16309.0                533                22.0
3  2018-01-01    16702.0                524                16.0
4  2018-01-01    17523.0                492                24.0

proportionUpsellTransactions  proportionUpsellrevenue  businessmonth  w.day  \
0                0.087640                0.038784                1        0
1                0.045000                0.019313                1        0
2                0.041276                0.018664                1        0
3                0.030534                0.009755                1        0
4                0.048780                0.023666                1        0

weeknum  trend.var  public_holiday  ph_weekend  ramadan
0         1         1              1.0         0.0         0
1         1         1              1.0         0.0         0
2         1         1              1.0         0.0         0
3         1         1              1.0         0.0         0
4         1         1              1.0         0.0         0

```

```

In [10]: # Getting the promotions data
path2 = '/Users/jingwenshi/VK-Lab Dropbox/Maria Shi/Data/new.GoJek.Promos.csv'
banners = pd.read_csv(path2)

# Converting 'businessdate' to datetime
banners['businessdate'] = pd.to_datetime(banners['businessdate'], format='%Y-%m-%d')

# Selecting specific columns from banners
bannersToMerge = banners[['fkstoreid', 'businessdate', 'banner.binary', 'go.banner.slide', 'banner.day']]

# Displaying the first few rows of bannersToMerge
print(bannersToMerge.head())

# Converting 'businessdate' in file_merged to datetime (if not already in this format)
file_merged['businessdate'] = pd.to_datetime(file_merged['businessdate'], format='%Y-%m-%d')

# Merging file_merged with bannersToMerge
file_merged = pd.merge(file_merged, bannersToMerge, on=['fkstoreid', 'businessdate'], how='left')

```

```
# Displaying the first few rows of the merged DataFrame
```

```
print(file_merged.head())
```

```

fkstoreid businessdate  banner.binary  go.banner.slide  banner.day
0      16054   2018-01-01           0           0           0
1      16054   2018-01-02           0           0           0
2      16054   2018-01-03           0           0           0
3      16054   2018-01-04           0           0           0
4      16054   2018-01-05           0           0           0
businessdate  fkstoreid  total_transactions  upsell_transactions  \
0   2018-01-01   16054.0             445             39.0
1   2018-01-01   16227.0             600             27.0
2   2018-01-01   16309.0             533             22.0
3   2018-01-01   16702.0             524             16.0
4   2018-01-01   17523.0             492             24.0

proportionUpsellTransactions  proportionUpsellrevenue  businessmonth  w.day  \
0              0.087640              0.038784              1          0
1              0.045000              0.019313              1          0
2              0.041276              0.018664              1          0
3              0.030534              0.009755              1          0
4              0.048780              0.023666              1          0

weeknum  trend.var  public_holiday  ph_weekend  ramadan  banner.binary  \
0         1         1              1.0         0.0         0           0.0
1         1         1              1.0         0.0         0           0.0
2         1         1              1.0         0.0         0           0.0
3         1         1              1.0         0.0         0           0.0
4         1         1              1.0         0.0         0           0.0

go.banner.slide  banner.day
0              0.0         0.0
1              0.0         0.0
2              0.0         0.0
3              0.0         0.0
4              0.0         0.0

```

```
In [11]: # Getting the dependence data
path3 = '/Users/jingwenshi/VK-Lab Dropbox/Maria Shi/Data/rstudio-export-20230725/2018.site.daily.platform.dependence.csv'
site_daily_dependence = pd.read_csv(path3)
```

```
# Housekeeping the Dependence Data
```

```
dependenceData = site_daily_dependence[['fkstoreid', 'businessdate', 'fourteen.day.full.price.sales.dep', 'fourteen.day.disc.price.sales.dep']]
```

```
# Converting 'businessdate' to datetime
```

```
dependenceData['businessdate'] = pd.to_datetime(dependenceData['businessdate'])
```

```
# Displaying the first few rows of dependenceData
```

```
print(dependenceData.head())
```



|   | fkstoreid | businessdate | fourteen.day.full.price.sales.dep | \   |
|---|-----------|--------------|-----------------------------------|-----|
| 0 | 16054     | 2018-01-01   |                                   | NaN |
| 1 | 16054     | 2018-01-02   |                                   | NaN |
| 2 | 16054     | 2018-01-03   |                                   | NaN |
| 3 | 16054     | 2018-01-04   |                                   | NaN |
| 4 | 16054     | 2018-01-05   |                                   | NaN |

|   | fourteen.day.disc.price.sales.dep |
|---|-----------------------------------|
| 0 | NaN                               |
| 1 | NaN                               |
| 2 | NaN                               |
| 3 | NaN                               |
| 4 | NaN                               |

/var/folders/y2/30zds30n5gv0yf5nk5xk2c9c0000gn/T/ipykernel\_20823/1009756989.py:9: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: [https://pandas.pydata.org/pandas-docs/stable/user\\_guide/indexing.html#returning-a-view-versus-a-copy](https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

dependenceData['businessdate'] = pd.to\_datetime(dependenceData['businessdate'])

```
In [12]: # Merging the 2 dataframes
file_merged = pd.merge(file_merged, dependenceData, on=['fkstoreid', 'businessdate'])

# Displaying the first few rows of the merged DataFrame
print(file_merged.head())
```

|   | businessdate | fkstoreid | total_transactions | upsell_transactions | \ |
|---|--------------|-----------|--------------------|---------------------|---|
| 0 | 2018-01-01   | 16054.0   | 445                | 39.0                |   |
| 1 | 2018-01-01   | 16227.0   | 600                | 27.0                |   |
| 2 | 2018-01-01   | 16309.0   | 533                | 22.0                |   |
| 3 | 2018-01-01   | 16702.0   | 524                | 16.0                |   |
| 4 | 2018-01-01   | 17523.0   | 492                | 24.0                |   |

|   | proportionUpsellTransactions | proportionUpsellrevenue | businessmonth | w.day | \ |
|---|------------------------------|-------------------------|---------------|-------|---|
| 0 | 0.087640                     | 0.038784                | 1             | 0     |   |
| 1 | 0.045000                     | 0.019313                | 1             | 0     |   |
| 2 | 0.041276                     | 0.018664                | 1             | 0     |   |
| 3 | 0.030534                     | 0.009755                | 1             | 0     |   |
| 4 | 0.048780                     | 0.023666                | 1             | 0     |   |

|   | weeknum | trend.var | public_holiday | ph_weekend | ramadan | banner.binary | \ |
|---|---------|-----------|----------------|------------|---------|---------------|---|
| 0 | 1       | 1         | 1.0            | 0.0        | 0       | 0.0           |   |
| 1 | 1       | 1         | 1.0            | 0.0        | 0       | 0.0           |   |
| 2 | 1       | 1         | 1.0            | 0.0        | 0       | 0.0           |   |
| 3 | 1       | 1         | 1.0            | 0.0        | 0       | 0.0           |   |
| 4 | 1       | 1         | 1.0            | 0.0        | 0       | 0.0           |   |

|   | go.banner.slide | banner.day | fourteen.day.full.price.sales.dep | \ |
|---|-----------------|------------|-----------------------------------|---|
| 0 | 0.0             | 0.0        | NaN                               |   |
| 1 | 0.0             | 0.0        | NaN                               |   |
| 2 | 0.0             | 0.0        | NaN                               |   |
| 3 | 0.0             | 0.0        | NaN                               |   |
| 4 | 0.0             | 0.0        | NaN                               |   |

|   | fourteen.day.disc.price.sales.dep |
|---|-----------------------------------|
| 0 | NaN                               |
| 1 | NaN                               |
| 2 | NaN                               |
| 3 | NaN                               |
| 4 | NaN                               |

```
In [13]: # Converting to categorical data
file_merged['fkstoreid'] = file_merged['fkstoreid'].astype('category')
file_merged['businessmonth'] = file_merged['businessmonth'].astype('category') # Assuming 'businessmonth' is in file_merged
file_merged['w.day'] = file_merged['w.day'].astype('category') # Assuming 'w.day' is in file_merged
file_merged['weeknum'] = file_merged['weeknum'].astype('category')

summary = file_merged.describe(include='all')

# Print the summary
print(summary)

# Number of Rows in the DataFrame
num_rows = len(file_merged)

# Print the number of rows
print(f"Number of Rows: {num_rows}")
```

|        |                     |           |                    |   |
|--------|---------------------|-----------|--------------------|---|
|        | businessdate        | fkstoreid | total_transactions | \ |
| count  | 34268               | 34268.0   | 34268.000000       |   |
| unique | 365                 | 99.0      | NaN                |   |
| top    | 2018-07-02 00:00:00 | 23329.0   | NaN                |   |
| freq   | 99                  | 365.0     | NaN                |   |
| first  | 2018-01-01 00:00:00 | NaN       | NaN                |   |
| last   | 2018-12-31 00:00:00 | NaN       | NaN                |   |
| mean   | NaN                 | NaN       | 426.798763         |   |
| std    | NaN                 | NaN       | 194.307546         |   |
| min    | NaN                 | NaN       | 17.000000          |   |
| 25%    | NaN                 | NaN       | 282.000000         |   |
| 50%    | NaN                 | NaN       | 394.000000         |   |
| 75%    | NaN                 | NaN       | 538.000000         |   |
| max    | NaN                 | NaN       | 2563.000000        |   |

|        |                     |                              |   |
|--------|---------------------|------------------------------|---|
|        | upsell_transactions | proportionUpsellTransactions | \ |
| count  | 34268.000000        | 34268.000000                 |   |
| unique | NaN                 | NaN                          |   |
| top    | NaN                 | NaN                          |   |
| freq   | NaN                 | NaN                          |   |
| first  | NaN                 | NaN                          |   |
| last   | NaN                 | NaN                          |   |
| mean   | 32.021273           | 0.076566                     |   |
| std    | 34.174655           | 0.070725                     |   |
| min    | 1.000000            | 0.001751                     |   |
| 25%    | 12.000000           | 0.033981                     |   |
| 50%    | 22.000000           | 0.056102                     |   |
| 75%    | 39.000000           | 0.091187                     |   |
| max    | 626.000000          | 0.661721                     |   |

|        |                         |               |         |         |   |
|--------|-------------------------|---------------|---------|---------|---|
|        | proportionUpsellrevenue | businessmonth | w.day   | weeknum | \ |
| count  | 34268.000000            | 34268.0       | 34268.0 | 34268.0 |   |
| unique | NaN                     | 12.0          | 7.0     | 5.0     |   |
| top    | NaN                     | 12.0          | 0.0     | 4.0     |   |
| freq   | NaN                     | 3069.0        | 4964.0  | 7905.0  |   |
| first  | NaN                     | NaN           | NaN     | NaN     |   |
| last   | NaN                     | NaN           | NaN     | NaN     |   |
| mean   | 0.034687                | NaN           | NaN     | NaN     |   |
| std    | 0.030889                | NaN           | NaN     | NaN     |   |
| min    | 0.000276                | NaN           | NaN     | NaN     |   |
| 25%    | 0.015457                | NaN           | NaN     | NaN     |   |
| 50%    | 0.025705                | NaN           | NaN     | NaN     |   |
| 75%    | 0.041842                | NaN           | NaN     | NaN     |   |
| max    | 0.254986                | NaN           | NaN     | NaN     |   |

|        |              |                |              |              |   |
|--------|--------------|----------------|--------------|--------------|---|
|        | trend.var    | public_holiday | ph_weekend   | ramadan      | \ |
| count  | 34268.000000 | 34268.000000   | 34268.000000 | 34268.000000 |   |
| unique | NaN          | NaN            | NaN          | NaN          |   |
| top    | NaN          | NaN            | NaN          | NaN          |   |
| freq   | NaN          | NaN            | NaN          | NaN          |   |
| first  | NaN          | NaN            | NaN          | NaN          |   |
| last   | NaN          | NaN            | NaN          | NaN          |   |
| mean   | 189.196335   | 0.04923        | 0.008113     | 0.082147     |   |
| std    | 104.118549   | 0.21635        | 0.089705     | 0.274592     |   |
| min    | 1.000000     | 0.00000        | 0.000000     | 0.000000     |   |

|     |            |         |          |          |
|-----|------------|---------|----------|----------|
| 25% | 101.000000 | 0.00000 | 0.000000 | 0.000000 |
| 50% | 192.000000 | 0.00000 | 0.000000 | 0.000000 |
| 75% | 279.000000 | 0.00000 | 0.000000 | 0.000000 |
| max | 365.000000 | 1.00000 | 1.000000 | 1.000000 |

|        | banner.binary | go.banner.slide | banner.day \ |
|--------|---------------|-----------------|--------------|
| count  | 34268.000000  | 34268.000000    | 34268.000000 |
| unique | NaN           | NaN             | NaN          |
| top    | NaN           | NaN             | NaN          |
| freq   | NaN           | NaN             | NaN          |
| first  | NaN           | NaN             | NaN          |
| last   | NaN           | NaN             | NaN          |
| mean   | 0.317702      | 0.458066        | 1.678096     |
| std    | 0.465590      | 0.900980        | 3.144231     |
| min    | 0.000000      | 0.000000        | 0.000000     |
| 25%    | 0.000000      | 0.000000        | 0.000000     |
| 50%    | 0.000000      | 0.000000        | 0.000000     |
| 75%    | 1.000000      | 1.000000        | 2.000000     |
| max    | 1.000000      | 5.000000        | 28.000000    |

|        | fourteen.day.full.price.sales.dep | fourteen.day.disc.price.sales.dep |
|--------|-----------------------------------|-----------------------------------|
| count  | 3.327000e+04                      | 3.327000e+04                      |
| unique | NaN                               | NaN                               |
| top    | NaN                               | NaN                               |
| freq   | NaN                               | NaN                               |
| first  | NaN                               | NaN                               |
| last   | NaN                               | NaN                               |
| mean   | 9.678813e-02                      | 1.007196e-01                      |
| std    | 6.569829e-02                      | 6.829340e-02                      |
| min    | -1.049062e-17                     | -1.049062e-17                     |
| 25%    | 4.668530e-02                      | 4.870581e-02                      |
| 50%    | 9.156814e-02                      | 9.574739e-02                      |
| 75%    | 1.399844e-01                      | 1.454965e-01                      |
| max    | 3.396912e-01                      | 3.541675e-01                      |

Number of Rows: 34268

/var/folders/y2/30zds30n5gv0yf5nk5xk2c9c0000gn/T/ipykernel\_20823/246247860.py:7: FutureWarning: Treating datetime data as categorical rather than numeric in `.describe` is deprecated and will be removed in a future version of pandas. Specify `datetime\_is\_numeric=True` to silence this warning and adopt the future behavior now.

```
summary = file_merged.describe(include='all')
```

## Logistic regression for %UpsellRevenue weighted by num.of transactions/day/store

```
In [14]: import statsmodels.api as sm
```

```
In [15]: # Selecting the features and the target variable
X = file_merged[['fkstoreid', 'w.day', 'businessmonth', 'weeknum', 'trend.var',
                 'public_holiday', 'ph_weekend', 'ramadan', 'banner.binary',
                 'go.banner.slide', 'banner.day', 'fourteen.day.disc.price.sales.dep']]
y = file_merged['proportionUpsellrevenue']

# Drop rows with missing values
X.dropna(inplace=True)
```

```

y = y.loc[X.index] # Make sure y has the same rows as X after dropping

# Add a constant to the model (intercept)
X = sm.add_constant(X)

# Weights
weights = file_merged.loc[X.index, 'total_transactions'] # Ensure weights align with X and y

# Fit the logistic regression model using GLM with a binomial family
model = sm.GLM(y, X, family=sm.families.Binomial(), weights=weights)
results = model.fit()

# Print the summary of the regression
print(results.summary())

```

```

=====
Generalized Linear Model Regression Results
=====
Dep. Variable:      proportionUpsellrevenue      No. Observations:      33270
Model:              GLM                        Df Residuals:          33257
Model Family:       Binomial                   Df Model:              12
Link Function:      Logit                      Scale:                 1.0000
Method:             IRLS                      Log-Likelihood:        -3814.3
Date:               Fri, 23 Feb 2024           Deviance:              524.08
Time:               10:15:04                   Pearson chi2:          607.
No. Iterations:     7                          Pseudo R-squ. (CS):    0.006084
Covariance Type:    nonrobust

=====

```

|                                   | coef       | std err  | z      | P> z  | [0.025    | 0.975]    |
|-----------------------------------|------------|----------|--------|-------|-----------|-----------|
| const                             | -2.3143    | 0.546    | -4.235 | 0.000 | -3.385    | -1.243    |
| fkstoreid                         | -5.231e-05 | 9.94e-06 | -5.260 | 0.000 | -7.18e-05 | -3.28e-05 |
| w.day                             | -0.0076    | 0.016    | -0.479 | 0.632 | -0.039    | 0.024     |
| businessmonth                     | -0.3628    | 0.445    | -0.816 | 0.415 | -1.234    | 0.509     |
| weeknum                           | -0.0954    | 0.101    | -0.943 | 0.345 | -0.294    | 0.103     |
| trend.var                         | 0.0159     | 0.015    | 1.088  | 0.277 | -0.013    | 0.045     |
| public_holiday                    | -0.0374    | 0.162    | -0.230 | 0.818 | -0.356    | 0.281     |
| ph_weekend                        | -0.3532    | 0.475    | -0.744 | 0.457 | -1.284    | 0.577     |
| ramadan                           | -0.2759    | 0.137    | -2.010 | 0.044 | -0.545    | -0.007    |
| banner.binary                     | -0.0943    | 0.138    | -0.682 | 0.495 | -0.366    | 0.177     |
| go.banner.slide                   | 0.0447     | 0.043    | 1.034  | 0.301 | -0.040    | 0.130     |
| banner.day                        | -0.0016    | 0.018    | -0.086 | 0.932 | -0.037    | 0.034     |
| fourteen.day.disc.price.sales.dep | -2.0622    | 0.451    | -4.571 | 0.000 | -2.946    | -1.178    |

```

=====

```

```

/var/folders/y2/30zds30n5gv0yf5nk5xk2c9c0000gn/T/ipykernel_20823/1017002940.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

```

```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
X.dropna(inplace=True)
/Users/jingwenshi/anaconda3/lib/python3.11/site-packages/statsmodels/base/model.py:130: ValueWarning: unknown kwargs ['weights']
warnings.warn(msg, ValueWarning)

```

Logistic regression for %UpsellOrder weighted by num.of transactions/day/store

```
In [16]: # Selecting the features and the target variable
X = file_merged[['fkstoreid', 'w.day', 'businessmonth', 'weeknum', 'trend.var',
                 'public_holiday', 'ph_weekend', 'ramadan', 'banner.binary',
                 'go.banner.slide', 'banner.day', 'fourteen.day.disc.price.sales.dep']]
y = file_merged['proportionUpsellTransactions']

# Drop rows with missing values
X.dropna(inplace=True)
y = y.loc[X.index] # Make sure y has the same rows as X after dropping

# Add a constant to the model (intercept)
X = sm.add_constant(X)

# Weights
weights = file_merged.loc[X.index, 'total_transactions'] # Ensure weights align with X and y

# Fit the logistic regression model using GLM with a binomial family
model = sm.GLM(y, X, family=sm.families.Binomial(), weights=weights)
results = model.fit()

# Print the summary of the regression
print(results.summary())
```

#### Generalized Linear Model Regression Results

|                  |                              |                     |          |
|------------------|------------------------------|---------------------|----------|
| Dep. Variable:   | proportionUpsellTransactions | No. Observations:   | 33270    |
| Model:           | GLM                          | Df Residuals:       | 33257    |
| Model Family:    | Binomial                     | Df Model:           | 12       |
| Link Function:   | Logit                        | Scale:              | 1.0000   |
| Method:          | IRLS                         | Log-Likelihood:     | -6544.6  |
| Date:            | Fri, 23 Feb 2024             | Deviance:           | 1272.4   |
| Time:            | 10:15:04                     | Pearson chi2:       | 1.48e+03 |
| No. Iterations:  | 6                            | Pseudo R-squ. (CS): | 0.01487  |
| Covariance Type: | nonrobust                    |                     |          |

  

|                                   | coef       | std err  | z      | P> z  | [0.025    | 0.975]    |
|-----------------------------------|------------|----------|--------|-------|-----------|-----------|
| const                             | -1.2334    | 0.377    | -3.268 | 0.001 | -1.973    | -0.494    |
| fkstoreid                         | -6.493e-05 | 6.88e-06 | -9.440 | 0.000 | -7.84e-05 | -5.14e-05 |
| w.day                             | 0.0013     | 0.011    | 0.114  | 0.909 | -0.020    | 0.023     |
| businessmonth                     | -0.4114    | 0.307    | -1.339 | 0.181 | -1.014    | 0.191     |
| weeknum                           | -0.0986    | 0.070    | -1.412 | 0.158 | -0.236    | 0.038     |
| trend.var                         | 0.0177     | 0.010    | 1.755  | 0.079 | -0.002    | 0.038     |
| public_holiday                    | -0.0074    | 0.111    | -0.067 | 0.947 | -0.225    | 0.210     |
| ph_weekend                        | -0.3416    | 0.315    | -1.085 | 0.278 | -0.959    | 0.275     |
| ramadan                           | -0.2671    | 0.093    | -2.863 | 0.004 | -0.450    | -0.084    |
| banner.binary                     | -0.1186    | 0.096    | -1.238 | 0.216 | -0.306    | 0.069     |
| go.banner.slide                   | 0.0576     | 0.030    | 1.915  | 0.055 | -0.001    | 0.117     |
| banner.day                        | 0.0005     | 0.013    | 0.042  | 0.967 | -0.024    | 0.025     |
| fourteen.day.disc.price.sales.dep | -2.2238    | 0.313    | -7.112 | 0.000 | -2.837    | -1.611    |

```
/var/folders/y2/30zds30n5gv0yf5nk5xk2c9c0000gn/T/ipykernel_20823/1704823843.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
X.dropna(inplace=True)
/Users/jingwenshi/anaconda3/lib/python3.11/site-packages/statsmodels/base/model.py:130: ValueWarning: unknown kwargs ['weights']
warnings.warn(msg, ValueWarning)
```

## Moderation Analysis

```
In [17]: import numpy as np
#import customer_feature_dataset

filepath = "/Users/jingwenshi/VK-Lab Dropbox/Maria Shi/Data/2018.site.cust.features.csv"
site_cust_features = pd.read_csv(filepath)
site_cust_features.head()
```

```
Out[17]:
```

|   | fkstoreid | avg.core.full.price | avg.core.disc.price | avg.core.per.check | median.core.full.price | median.core.disc.price | median.core.per.check | num.checks | num.days | peak.checks | ... | bin.n |
|---|-----------|---------------------|---------------------|--------------------|------------------------|------------------------|-----------------------|------------|----------|-------------|-----|-------|
| 0 | 16054     | 21752.783433        | 17005.051154        | 1.648246           | 21590.5                | 15264.0                | 1                     | 146958     | 349      | 86578       | ... |       |
| 1 | 16227     | 21878.763957        | 16942.756175        | 1.981727           | 22500.0                | 15297.0                | 1                     | 207742     | 365      | 110425      | ... |       |
| 2 | 16309     | 20353.936036        | 15878.789671        | 1.492275           | 20909.0                | 15000.0                | 1                     | 170106     | 365      | 103006      | ... |       |
| 3 | 16702     | 22320.903108        | 17381.762269        | 1.859957           | 22500.0                | 15297.0                | 1                     | 211314     | 365      | 124353      | ... |       |
| 4 | 17523     | 19216.920504        | 14511.912515        | 1.681073           | 20909.0                | 14345.0                | 1                     | 147087     | 365      | 82787       | ... |       |

5 rows × 34 columns

```
In [18]: ## housekeeping
# create new boolean variable: "lunch.store"
site_cust_features['lunch.store'] = site_cust_features['lunch.peak.share'] > 0.5

# log(lunch.peak.share)
site_cust_features['lunch.ratio'] = np.log(site_cust_features['lunch.peak.share'] / (1 - site_cust_features['lunch.peak.share']))

# shrink (avg.core.full.price)
site_cust_features['avg.core.full.price'] = site_cust_features['avg.core.full.price'] / 10000

## Standardisation
# def a standardisation function
def standardize(series):
    return (series - series.mean()) / series.std()

columns_to_standardize = [
    'bin.avg.core.full.price', 'bin.avg.core.per.check', 'lunch.ratio',
    'avg.core.full.price', 'avg.core.per.check', 'bin.median.core.full.price',
    'bin.median.core.per.check', 'median.core.full.price', 'median.core.per.check'
]

for col in columns_to_standardize:
```

```
standardized_col_name = f'std.{col}'
site_cust_features[standardized_col_name] = standardize(site_cust_features[col])
```

```
In [19]: # Merge the two dataframes together
to_regress = pd.merge(file_merged, site_cust_features, on='fkstoreid', how='left')
```

```
# Print the number of rows
print(f"Number of Rows: {len(to_regress)}")
```

```
print(to_regress.columns)
```

Number of Rows: 34268

```
Index(['businessdate', 'fkstoreid', 'total_transactions',
      'upsell_transactions', 'proportionUpsellTransactions',
      'proportionUpsellrevenue', 'businessmonth', 'w.day', 'weeknum',
      'trend.var', 'public_holiday', 'ph_weekend', 'ramadan', 'banner.binary',
      'go.banner.slide', 'banner.day', 'fourteen.day.full.price.sales.dep',
      'fourteen.day.disc.price.sales.dep', 'avg.core.full.price',
      'avg.core.disc.price', 'avg.core.per.check', 'median.core.full.price',
      'median.core.disc.price', 'median.core.per.check', 'num.checks',
      'num.days', 'peak.checks', 'lunch.peak.share',
      'ofln.avg.core.full.price', 'ofln.avg.core.disc.price',
      'ofln.avg.core.per.check', 'ofln.median.core.full.price',
      'ofln.median.core.disc.price', 'ofln.median.core.per.check',
      'ofln.num.checks', 'ofln.num.days', 'avg.daily.checks',
      'bin.avg.core.full.price', 'bin.avg.core.disc.price',
      'bin.avg.core.per.check', 'bin.median.core.full.price',
      'bin.median.core.disc.price', 'bin.median.core.per.check',
      'bin.avg.daily.checks', 'ofln.avg.daily.checks',
      'ofln.bin.avg.core.full.price', 'ofln.bin.avg.core.per.check',
      'dec.avg.core.full.price', 'qrt.avg.core.full.price',
      'tri.avg.core.full.price', 'tri.avg.core.per.check', 'lunch.store',
      'lunch.ratio', 'std.bin.avg.core.full.price',
      'std.bin.avg.core.per.check', 'std.lunch.ratio',
      'std.avg.core.full.price', 'std.avg.core.per.check',
      'std.bin.median.core.full.price', 'std.bin.median.core.per.check',
      'std.median.core.full.price', 'std.median.core.per.check'],
      dtype='object')
```

```
In [20]: # Inserting the lag variable
to_regress.sort_values(by=['fkstoreid', 'trend.var'], inplace=True)
to_regress['lag_daily_upsell_checks_prop'] = to_regress.groupby('fkstoreid')['proportionUpsellTransactions'].shift(1)
to_regress['lag_daily_upsell_revenue_prop'] = to_regress.groupby('fkstoreid')['proportionUpsellrevenue'].shift(1)
```

```
In [21]: import statsmodels.api as sm
import statsmodels.formula.api as smf
from patsy import dmatrices

# Define the base and extra parts of the model formula
LHS = 'proportionUpsellrevenue'
RHS_base = ' + '.join(['Q("banner.binary")', 'Q("w.day")', 'Q("businessmonth")', 'Q("weeknum")',
                       'Q("public_holiday")', 'Q("ramadan")', 'Q("trend.var")'])

RHS_extra = [
```



```

'Q("fourteen.day.full.price.sales.dep")',
'Q("fourteen.day.full.price.sales.dep") + Q("fourteen.day.full.price.sales.dep"):Q("std.avg.core.full.price")',
'Q("fourteen.day.full.price.sales.dep") + Q("fourteen.day.full.price.sales.dep"):Q("std.avg.core.per.check")',
'Q("fourteen.day.full.price.sales.dep") + Q("fourteen.day.full.price.sales.dep"):Q("std.lunch.ratio")'
]

# Combine the RHS parts and create the full formula
formulae = [f"{LHS} ~ {extra} + {RHS_base}" for extra in RHS_extra] # Removed | fkstoreid because it's not directly supported in formula

# Prepare data
filtered_data = to_regress[to_regress['proportionUpsellTransactions'].notna()]

# Run the models
models = []
for formula_str in formulae:
    model = smf.glm(formula=formula_str, data=filtered_data, family=sm.families.Binomial()).fit()
    models.append(model)

# Custom output formatting
for model in models:
    print(model.summary())

```

## Generalized Linear Model Regression Results

```

=====
Dep. Variable:    proportionUpsellrevenue    No. Observations:    33270
Model:            GLM                      Df Residuals:        33243
Model Family:     Binomial                 Df Model:            26
Link Function:     Logit                   Scale:               1.0000
Method:           IRLS                    Log-Likelihood:      -3813.5
Date:             Fri, 23 Feb 2024         Deviance:            522.38
Time:             10:15:05                 Pearson chi2:         637.
No. Iterations:   7                       Pseudo R-squ. (CS):  0.006135
Covariance Type:  nonrobust
=====

```

|                                        | coef    | std err | z       | P> z  | [0.025 | 0.975] |
|----------------------------------------|---------|---------|---------|-------|--------|--------|
| Intercept                              | -3.4068 | 0.179   | -19.068 | 0.000 | -3.757 | -3.057 |
| Q("w.day") [T.1]                       | -0.0245 | 0.114   | -0.215  | 0.830 | -0.248 | 0.199  |
| Q("w.day") [T.2]                       | -0.0381 | 0.113   | -0.339  | 0.735 | -0.259 | 0.183  |
| Q("w.day") [T.3]                       | -0.0340 | 0.113   | -0.302  | 0.763 | -0.255 | 0.187  |
| Q("w.day") [T.4]                       | -0.0438 | 0.114   | -0.383  | 0.702 | -0.268 | 0.180  |
| Q("w.day") [T.5]                       | -0.0512 | 0.112   | -0.456  | 0.648 | -0.271 | 0.169  |
| Q("w.day") [T.6]                       | -0.0719 | 0.113   | -0.639  | 0.523 | -0.292 | 0.149  |
| Q("businessmonth") [T.2]               | -0.1098 | 0.525   | -0.209  | 0.834 | -1.140 | 0.920  |
| Q("businessmonth") [T.3]               | -0.2122 | 0.956   | -0.222  | 0.824 | -2.085 | 1.661  |
| Q("businessmonth") [T.4]               | -0.1453 | 1.442   | -0.101  | 0.920 | -2.971 | 2.681  |
| Q("businessmonth") [T.5]               | -0.1077 | 1.917   | -0.056  | 0.955 | -3.865 | 3.650  |
| Q("businessmonth") [T.6]               | -0.3089 | 2.409   | -0.128  | 0.898 | -5.030 | 4.412  |
| Q("businessmonth") [T.7]               | 0.0306  | 2.884   | 0.011   | 0.992 | -5.623 | 5.684  |
| Q("businessmonth") [T.8]               | 0.1869  | 3.377   | 0.055   | 0.956 | -6.432 | 6.806  |
| Q("businessmonth") [T.9]               | 0.4844  | 3.869   | 0.125   | 0.900 | -7.099 | 8.068  |
| Q("businessmonth") [T.10]              | 0.5802  | 4.346   | 0.134   | 0.894 | -7.938 | 9.098  |
| Q("businessmonth") [T.11]              | 0.8687  | 4.839   | 0.180   | 0.858 | -8.616 | 10.353 |
| Q("businessmonth") [T.12]              | 0.9803  | 5.314   | 0.184   | 0.854 | -9.435 | 11.395 |
| Q("weeknum") [T.2]                     | 0.0305  | 0.143   | 0.213   | 0.831 | -0.251 | 0.312  |
| Q("weeknum") [T.3]                     | 0.0501  | 0.240   | 0.209   | 0.834 | -0.419 | 0.520  |
| Q("weeknum") [T.4]                     | 0.0612  | 0.343   | 0.178   | 0.858 | -0.611 | 0.734  |
| Q("weeknum") [T.5]                     | 0.0428  | 0.431   | 0.099   | 0.921 | -0.801 | 0.887  |
| Q("fourteen.day.full.price.sales.dep") | -2.4845 | 0.479   | -5.186  | 0.000 | -3.423 | -1.546 |
| Q("banner.binary")                     | -0.0619 | 0.072   | -0.857  | 0.391 | -0.203 | 0.080  |
| Q("public_holiday")                    | -0.0056 | 0.162   | -0.035  | 0.972 | -0.323 | 0.311  |
| Q("ramadan")                           | 0.0137  | 0.184   | 0.074   | 0.941 | -0.347 | 0.374  |
| Q("trend.var")                         | 0.0003  | 0.016   | 0.017   | 0.986 | -0.031 | 0.031  |

## Generalized Linear Model Regression Results

```

=====
Dep. Variable:    proportionUpsellrevenue    No. Observations:    33270
Model:            GLM                      Df Residuals:        33242
Model Family:     Binomial                 Df Model:            27
Link Function:     Logit                   Scale:               1.0000
Method:           IRLS                    Log-Likelihood:      -3780.5
Date:             Fri, 23 Feb 2024         Deviance:            456.48
Time:             10:15:05                 Pearson chi2:         527.
No. Iterations:   7                       Pseudo R-squ. (CS):  0.008102
Covariance Type:  nonrobust
=====

```

```

=====
coef      std err      z      P>|z|      [0.025      0.975]
=====

```

|                                                                     |         |       |         |       |        |        |
|---------------------------------------------------------------------|---------|-------|---------|-------|--------|--------|
| Intercept                                                           | -3.3683 | 0.179 | -18.841 | 0.000 | -3.719 | -3.018 |
| Q("w.day") [T.1]                                                    | -0.0247 | 0.114 | -0.216  | 0.829 | -0.249 | 0.199  |
| Q("w.day") [T.2]                                                    | -0.0386 | 0.113 | -0.342  | 0.732 | -0.260 | 0.182  |
| Q("w.day") [T.3]                                                    | -0.0336 | 0.113 | -0.298  | 0.765 | -0.255 | 0.187  |
| Q("w.day") [T.4]                                                    | -0.0432 | 0.114 | -0.378  | 0.706 | -0.267 | 0.181  |
| Q("w.day") [T.5]                                                    | -0.0506 | 0.112 | -0.450  | 0.653 | -0.271 | 0.170  |
| Q("w.day") [T.6]                                                    | -0.0717 | 0.113 | -0.636  | 0.525 | -0.293 | 0.149  |
| Q("businessmonth") [T.2]                                            | -0.1170 | 0.526 | -0.222  | 0.824 | -1.148 | 0.914  |
| Q("businessmonth") [T.3]                                            | -0.2118 | 0.957 | -0.221  | 0.825 | -2.087 | 1.664  |
| Q("businessmonth") [T.4]                                            | -0.1391 | 1.444 | -0.096  | 0.923 | -2.969 | 2.691  |
| Q("businessmonth") [T.5]                                            | -0.0872 | 1.920 | -0.045  | 0.964 | -3.850 | 3.676  |
| Q("businessmonth") [T.6]                                            | -0.3053 | 2.413 | -0.127  | 0.899 | -5.034 | 4.423  |
| Q("businessmonth") [T.7]                                            | 0.0366  | 2.889 | 0.013   | 0.990 | -5.625 | 5.699  |
| Q("businessmonth") [T.8]                                            | 0.1955  | 3.382 | 0.058   | 0.954 | -6.433 | 6.824  |
| Q("businessmonth") [T.9]                                            | 0.5113  | 3.875 | 0.132   | 0.895 | -7.084 | 8.107  |
| Q("businessmonth") [T.10]                                           | 0.6125  | 4.353 | 0.141   | 0.888 | -7.919 | 9.144  |
| Q("businessmonth") [T.11]                                           | 0.8983  | 4.847 | 0.185   | 0.853 | -8.601 | 10.398 |
| Q("businessmonth") [T.12]                                           | 1.0084  | 5.322 | 0.189   | 0.850 | -9.423 | 11.439 |
| Q("weeknum") [T.2]                                                  | 0.0341  | 0.144 | 0.237   | 0.812 | -0.247 | 0.316  |
| Q("weeknum") [T.3]                                                  | 0.0522  | 0.240 | 0.218   | 0.828 | -0.418 | 0.523  |
| Q("weeknum") [T.4]                                                  | 0.0600  | 0.344 | 0.175   | 0.861 | -0.614 | 0.734  |
| Q("weeknum") [T.5]                                                  | 0.0407  | 0.431 | 0.094   | 0.925 | -0.805 | 0.886  |
| Q("fourteen.day.full.price.sales.dep")                              | -3.0937 | 0.480 | -6.451  | 0.000 | -4.034 | -2.154 |
| Q("fourteen.day.full.price.sales.dep"):Q("std.avg.core.full.price") | 1.7637  | 0.193 | 9.142   | 0.000 | 1.386  | 2.142  |
| Q("banner.binary")                                                  | -0.0502 | 0.072 | -0.694  | 0.488 | -0.192 | 0.092  |
| Q("public_holiday")                                                 | -0.0054 | 0.162 | -0.034  | 0.973 | -0.323 | 0.312  |
| Q("ramadan")                                                        | 0.0185  | 0.184 | 0.101   | 0.920 | -0.342 | 0.380  |
| Q("trend.var")                                                      | 0.0003  | 0.016 | 0.017   | 0.986 | -0.031 | 0.032  |

#### Generalized Linear Model Regression Results

```

=====
Dep. Variable:    proportionUpsellrevenue    No. Observations:    33270
Model:            GLM                      Df Residuals:        33242
Model Family:     Binomial                 Df Model:             27
Link Function:    Logit                   Scale:               1.0000
Method:           IRLS                    Log-Likelihood:       -3813.4
Date:             Fri, 23 Feb 2024         Deviance:             522.22
Time:             10:15:05                 Pearson chi2:         637.
No. Iterations:   7                       Pseudo R-squ. (CS):   0.006140
Covariance Type:  nonrobust
=====

```

|                          | coef    | std err | z       | P> z  | [0.025 | 0.975] |
|--------------------------|---------|---------|---------|-------|--------|--------|
| Intercept                | -3.4090 | 0.179   | -19.071 | 0.000 | -3.759 | -3.059 |
| Q("w.day") [T.1]         | -0.0245 | 0.114   | -0.215  | 0.830 | -0.248 | 0.199  |
| Q("w.day") [T.2]         | -0.0381 | 0.113   | -0.338  | 0.735 | -0.259 | 0.183  |
| Q("w.day") [T.3]         | -0.0340 | 0.113   | -0.302  | 0.762 | -0.255 | 0.187  |
| Q("w.day") [T.4]         | -0.0438 | 0.114   | -0.384  | 0.701 | -0.268 | 0.180  |
| Q("w.day") [T.5]         | -0.0513 | 0.112   | -0.457  | 0.648 | -0.271 | 0.169  |
| Q("w.day") [T.6]         | -0.0720 | 0.113   | -0.639  | 0.523 | -0.293 | 0.149  |
| Q("businessmonth") [T.2] | -0.1095 | 0.525   | -0.208  | 0.835 | -1.139 | 0.920  |
| Q("businessmonth") [T.3] | -0.2123 | 0.956   | -0.222  | 0.824 | -2.085 | 1.661  |
| Q("businessmonth") [T.4] | -0.1456 | 1.442   | -0.101  | 0.920 | -2.972 | 2.680  |
| Q("businessmonth") [T.5] | -0.1093 | 1.917   | -0.057  | 0.955 | -3.867 | 3.648  |

|                                                                    |         |       |        |       |        |        |
|--------------------------------------------------------------------|---------|-------|--------|-------|--------|--------|
| Q("businessmonth") [T.6]                                           | -0.3076 | 2.409 | -0.128 | 0.898 | -5.029 | 4.414  |
| Q("businessmonth") [T.7]                                           | 0.0324  | 2.884 | 0.011  | 0.991 | -5.621 | 5.686  |
| Q("businessmonth") [T.8]                                           | 0.1888  | 3.377 | 0.056  | 0.955 | -6.430 | 6.808  |
| Q("businessmonth") [T.9]                                           | 0.4851  | 3.869 | 0.125  | 0.900 | -7.099 | 8.069  |
| Q("businessmonth") [T.10]                                          | 0.5805  | 4.346 | 0.134  | 0.894 | -7.937 | 9.098  |
| Q("businessmonth") [T.11]                                          | 0.8691  | 4.839 | 0.180  | 0.857 | -8.616 | 10.354 |
| Q("businessmonth") [T.12]                                          | 0.9806  | 5.314 | 0.185  | 0.854 | -9.435 | 11.396 |
| Q("weeknum") [T.2]                                                 | 0.0303  | 0.143 | 0.211  | 0.833 | -0.251 | 0.311  |
| Q("weeknum") [T.3]                                                 | 0.0501  | 0.240 | 0.209  | 0.834 | -0.420 | 0.520  |
| Q("weeknum") [T.4]                                                 | 0.0615  | 0.343 | 0.179  | 0.858 | -0.611 | 0.734  |
| Q("weeknum") [T.5]                                                 | 0.0431  | 0.431 | 0.100  | 0.920 | -0.801 | 0.887  |
| Q("fourteen.day.full.price.sales.dep")                             | -2.4184 | 0.507 | -4.769 | 0.000 | -3.412 | -1.425 |
| Q("fourteen.day.full.price.sales.dep"):Q("std.avg.core.per.check") | -0.1101 | 0.276 | -0.399 | 0.690 | -0.652 | 0.431  |
| Q("banner.binary")                                                 | -0.0627 | 0.072 | -0.868 | 0.385 | -0.204 | 0.079  |
| Q("public_holiday")                                                | -0.0055 | 0.162 | -0.034 | 0.973 | -0.323 | 0.312  |
| Q("ramadan")                                                       | 0.0133  | 0.184 | 0.072  | 0.942 | -0.347 | 0.374  |
| Q("trend.var")                                                     | 0.0003  | 0.016 | 0.017  | 0.987 | -0.031 | 0.031  |

# Generalized Linear Model Regression Results

|                  |                         |                     |          |
|------------------|-------------------------|---------------------|----------|
| Dep. Variable:   | proportionUpsellrevenue | No. Observations:   | 33270    |
| Model:           | GLM                     | Df Residuals:       | 33242    |
| Model Family:    | Binomial                | Df Model:           | 27       |
| Link Function:   | Logit                   | Scale:              | 1.0000   |
| Method:          | IRLS                    | Log-Likelihood:     | -3811.1  |
| Date:            | Fri, 23 Feb 2024        | Deviance:           | 517.62   |
| Time:            | 10:15:05                | Pearson chi2:       | 627.     |
| No. Iterations:  | 7                       | Pseudo R-squ. (CS): | 0.006277 |
| Covariance Type: | nonrobust               |                     |          |

|                                        | coef    | std err | z       | P> z  | [0.025 | 0.975] |
|----------------------------------------|---------|---------|---------|-------|--------|--------|
| Intercept                              | -3.3944 | 0.179   | -18.990 | 0.000 | -3.745 | -3.044 |
| Q("w.day") [T.1]                       | -0.0246 | 0.114   | -0.216  | 0.829 | -0.248 | 0.199  |
| Q("w.day") [T.2]                       | -0.0383 | 0.113   | -0.340  | 0.734 | -0.259 | 0.182  |
| Q("w.day") [T.3]                       | -0.0340 | 0.113   | -0.302  | 0.763 | -0.255 | 0.187  |
| Q("w.day") [T.4]                       | -0.0436 | 0.114   | -0.381  | 0.703 | -0.267 | 0.180  |
| Q("w.day") [T.5]                       | -0.0512 | 0.112   | -0.456  | 0.649 | -0.271 | 0.169  |
| Q("w.day") [T.6]                       | -0.0718 | 0.113   | -0.638  | 0.524 | -0.292 | 0.149  |
| Q("businessmonth") [T.2]               | -0.1111 | 0.525   | -0.211  | 0.833 | -1.141 | 0.919  |
| Q("businessmonth") [T.3]               | -0.2117 | 0.956   | -0.221  | 0.825 | -2.085 | 1.661  |
| Q("businessmonth") [T.4]               | -0.1452 | 1.442   | -0.101  | 0.920 | -2.971 | 2.681  |
| Q("businessmonth") [T.5]               | -0.1045 | 1.917   | -0.055  | 0.957 | -3.862 | 3.653  |
| Q("businessmonth") [T.6]               | -0.3152 | 2.409   | -0.131  | 0.896 | -5.037 | 4.407  |
| Q("businessmonth") [T.7]               | 0.0217  | 2.885   | 0.008   | 0.994 | -5.632 | 5.676  |
| Q("businessmonth") [T.8]               | 0.1779  | 3.377   | 0.053   | 0.958 | -6.441 | 6.797  |
| Q("businessmonth") [T.9]               | 0.4795  | 3.870   | 0.124   | 0.901 | -7.105 | 8.064  |
| Q("businessmonth") [T.10]              | 0.5773  | 4.346   | 0.133   | 0.894 | -7.941 | 9.096  |
| Q("businessmonth") [T.11]              | 0.8658  | 4.840   | 0.179   | 0.858 | -8.620 | 10.351 |
| Q("businessmonth") [T.12]              | 0.9775  | 5.314   | 0.184   | 0.854 | -9.439 | 11.394 |
| Q("weeknum") [T.2]                     | 0.0315  | 0.143   | 0.220   | 0.826 | -0.250 | 0.313  |
| Q("weeknum") [T.3]                     | 0.0504  | 0.240   | 0.210   | 0.834 | -0.419 | 0.520  |
| Q("weeknum") [T.4]                     | 0.0601  | 0.343   | 0.175   | 0.861 | -0.613 | 0.733  |
| Q("weeknum") [T.5]                     | 0.0415  | 0.431   | 0.096   | 0.923 | -0.803 | 0.886  |
| Q("fourteen.day.full.price.sales.dep") | -2.8165 | 0.503   | -5.604  | 0.000 | -3.802 | -1.831 |

|                                                             |         |       |        |       |        |        |
|-------------------------------------------------------------|---------|-------|--------|-------|--------|--------|
| Q("fourteen.day.full.price.sales.dep"):Q("std.lunch.ratio") | -0.5684 | 0.261 | -2.175 | 0.030 | -1.081 | -0.056 |
| Q("banner.binary")                                          | -0.0584 | 0.072 | -0.809 | 0.419 | -0.200 | 0.083  |
| Q("public_holiday")                                         | -0.0063 | 0.162 | -0.039 | 0.969 | -0.323 | 0.311  |
| Q("ramadan")                                                | 0.0160  | 0.184 | 0.087  | 0.931 | -0.345 | 0.377  |
| Q("trend.var")                                              | 0.0003  | 0.016 | 0.021  | 0.984 | -0.031 | 0.032  |
| =====                                                       |         |       |        |       |        |        |

In [ ]: